THE JACOBI-ZARISKI SEQUENCE OF A FORMALLY SMOOTH LOCAL HOMOMORPHISM AT THE RESIDUE FIELD

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Allowed standard prerequisites (editorial).

- Identification of the cotangent space with H_1 of the naive cotangent complex, and exactness of the Jacobi-Zariski sequence. Algebra, Lemma \ref{algebra-lemma-NL-surjection} and Algebra, Lemma \ref{algebra-lemma-exact-sequence-NL}, cited at proof lines 10029-10031. Reduce the claim to injectivity of H_1(NL_{K/A}) into m_B/m_B^2.
- The conormal sequence for a regular quotient and the criterion for a finite module to be cut by a regular sequence. Lemma \ref{lemma-conormal-sequence-H1-regular} and Lemma \ref{lemma-noetherian-finite-all-equivalent}, cited at proof lines 10048-10050. Produce the short exact sequence 0 to m_A/m_A^2 tensor K to m_B/m_B^2 to m_{B-bar}/m_{B-bar}^2 to 0.
- Formal smoothness of a local map implies flatness, and formal smoothness passes to the special fibre. Lemma \ref{lemma-formally-smooth-flat} and Lemma \ref{lemma-base-change-fs}, cited at proof lines 10052 and 10072. Give flatness of A to B and formal smoothness of the fibre extension, which is what makes the left vertical arrow of the diagram injective.
- Injectivity of the map on differentials induced by a formally smooth fibre extension. Theorem \ref{theorem-regular-fs}, cited at proof line 10071. Supplies the injectivity of the left vertical arrow, after which the snake lemma completes the proof.

Source reference. Original lemma 0DYH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17467–17467. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07BP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9349–9349. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00S2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1024–1024. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0666: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7925–7925. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09CC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11825–11825. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9769–9769. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9451–9451. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9446–9446. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Assume $A \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology. Let K be the residue field of B. Then the Jacobi-Zariski sequence for $A \to B \to K$ gives an exact sequence*

$$0 \to H_1(NL_{K/A}) \to \mathfrak{m}_B/\mathfrak{m}_B^2 \to \Omega_{B/A} \otimes_B K \to \Omega_{K/A} \to 0$$

Proof. Observe that $\mathfrak{m}_B/\mathfrak{m}_B^2 = H_1(NL_{K/B})$ by Algebra, Lemma 07BP. By Algebra, Lemma 00S2 it remains to show injectivity of $H_1(NL_{K/A}) \to \mathfrak{m}_B/\mathfrak{m}_B^2$. With k the residue field of A, the Jacobi-Zariski sequence for $A \to k \to K$ gives $\Omega_{K/A} = \Omega_{K/k}$ and an exact sequence

$$\mathfrak{m}_A/\mathfrak{m}_A^2 \otimes_k K \to H_1(NL_{K/A}) \to H_1(NL_{K/k}) \to 0$$

Set $\overline{B} = B \otimes_A k$. Since $\overline{B}$ is regular the ideal $\mathfrak{m}_{\overline{B}}$ is generated by a regular sequence. Applying Lemmas 0666 and 09CC to $\mathfrak{m}_A B \subset \mathfrak{m}_B$ we find $\mathfrak{m}_A B/(\mathfrak{m}_A B \cap \mathfrak{m}_B^2) = \mathfrak{m}_A B/\mathfrak{m}_A\mathfrak{m}_B$ which is equal to $\mathfrak{m}_A/\mathfrak{m}_A^2 \otimes_k K$ as $A \to B$ is flat by Lemma 07NP. Thus we obtain a short exact sequence

$$0 \to \mathfrak{m}_A/\mathfrak{m}_A^2 \otimes_k K \to \mathfrak{m}_B/\mathfrak{m}_B^2 \to \mathfrak{m}_{\overline{B}}/\mathfrak{m}_{\overline{B}}^2 \to 0$$

Functoriality of the Jacobi-Zariski sequences shows that we obtain a commutative diagram

$$\begin{array}{ccccccc} & & \mathfrak{m}_A/\mathfrak{m}_A^2 \otimes_k K & \longrightarrow & H_1(NL_{K/A}) & \longrightarrow & H_1(NL_{K/k}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 \longrightarrow & \mathfrak{m}_A/\mathfrak{m}_A^2 \otimes_k K & \longrightarrow & \mathfrak{m}_B/\mathfrak{m}_B^2 & \longrightarrow & \mathfrak{m}_{\overline{B}}/\mathfrak{m}_{\overline{B}}^2 & \longrightarrow 0 \end{array}$$

The left vertical arrow is injective by Theorem 07EL as $k \to \overline{B}$ is formally smooth in the $\mathfrak{m}_{\overline{B}}$-adic topology by Lemma 07EG. This finishes the proof by the snake lemma. □